SnapLedger — Analytics Event & Diagnostics Catalogue

Every Firebase Analytics event, user property, and Crashlytics non-fatal currently emitted by the iOS app, including the Analytics Advancement Brief (18 October 2026).

Generated: 18 October 2026

Source of truth: `src/Xpnse/Firebase/AppAnalytics.swift`, `AppDiagnostics.swift`, `AnalyticsUserProperties.swift`, `ActivationAnalytics.swift`, `TransactionFormAbandonment.swift`

Specs: Instrumentation Specification v1.0 + Advancement Brief v1.0

P0 required first P1 next P2 follow-up EXISTING pre-spec event, kept as is

Rules that apply to every event

- Outcome events fire only after the state change actually succeeds (for example `txn_save` only after persistence returns success).
- All parameter values are bounded enums or coarse buckets. No amounts, balances, merchant names, titles, notes, category names, OCR text, search queries, file names, error descriptions, or unique IDs are sent.
- Events that could repeat on SwiftUI redraws are guarded (once per form presentation, per visit, per session, or per state transition).
- Debug builds print only the event name and parameter *keys* to the console, never values.

1. Shared parameter values

Parameter	Allowed values
`source` (transactions)	`home`, `education_income`, `education_expense`, `scan`, `share_text`, `share_image`, `widget_add`, `widget_balance`, `notification`, `insights_suggestion`, `edit`
`txn_type`	`expense`, `income`, `savings`
`mode`	`create`, `edit`, `recurring_create`
`result`	`success`, `failed`, `cancelled`; event-specific values noted per event
`failure_reason`	`validation`, `persistence`, `permission_denied`, `unsupported_format`, `unavailable`, `unexpected`, `form_closed`, `no_text_found`, `invalid_image`
`journey`	`income_setup`, `expense_setup`, `recurring_reminder`, `custom_category`
`state`	`empty`, `ready`, `error`
`txn_bucket`	`0`, `1`, `2_5`, `6_20`, `21_plus`
`record_count_bucket`	`0`, `1_10`, `11_100`, `101_1000`, `1001_plus`
`duration_bucket` (startup)	`lt_1s`, `1_3s`, `3_10s`, `gt_10s`
`duration_bucket` (**`slow_operation`**)	`2_3s`, `3_10s`, `gt_10s` (trigger is ≥ 2s)
`hours_since_install_bucket`	`lt_1h`, `1_24h`, `1_3d`, `gt_3d`
`stage` (form)	`empty`, `partially_filled`, `ready_to_save`

exit_reason	`explicit_close` (not used — that is cancelled), `background_timeout`, `navigation`
Booleans (**recurring**, **enabled**, …)	`true`, `false`; `reminder` also allows `na`

2. Transaction lifecycle

Event	When it fires	Parameters
transaction_form_opened P0	Add / edit form is presented. Once per presentation.	`txn_type`, `source`, `mode`
transaction_validation_failed P0	An inline error blocks saving. Once per field per form.	`field`: `end_date`, `reminder_time`; `reason`: `invalid_range`, `invalid_schedule`; `txn_type`, `source`
txn_save P0	After a transaction or recurring rule is persisted successfully. Also used for Insights "Make recurring".	`txn_type`, `source`, `mode`, `recurring`, `reminder` (`true` / `false` / `na`)
first_transaction_saved P0	First successful non-edit save for this install / analytics identity. Once; not reset by Clear Local Data. Reinstall (UserDefaults gone) can emit again.	`txn_type`, `source`, `hours_since_install_bucket`
transaction_save_failed P0	Persistence returns failure.	`txn_type`, `source`, `failure_reason` = `persistence`
transaction_form_cancelled P0	User taps the close button. Not logged on background. Never combined with abandoned for the same exit.	`txn_type`, `source`, `mode`
transaction_form_abandoned P0	Inferred unfinished form. Backgrounding stores a marker only. Logged on return after 30 minutes, on next launch if the process died, or when the form leaves by navigation. At most once per presentation. No field values.	`txn_type`, `source`, `mode`, `stage`, `exit_reason`: `background_timeout`, `navigation`
transaction_deleted P1	User confirms delete and deletion succeeds.	`txn_type`, `source`
category_changed P2	Successful edit save where the category changed.	`change_source` = `edit_form`

Covered by `txn_save` instead of separate events (as the spec permits): `transaction_edited` → `mode=edit`; `bill_scan_saved` → `source=scan`; `recurring_transaction_created` → `mode=recurring_create`; `meaningful_action` → dedicated events.

3. Education journeys

Event	When it fires	Parameters
education_presented P0	Education sheet is actually shown (not on eligibility).	`journey`, `trigger`: `home_visit`, `recurring_save`

`education_action_selected` P0	User taps the primary action.	`journey`, `action`: `open_add_transaction`, `open_manage_categories`, `request_notification_permission`
`education_dismissed` P0	User closes the sheet without doing the task.	`journey`, `dismiss_method`: `secondary_button`, `swipe`
`education_completed` P0	Task actually completes (validated). First transition only.	`journey`
`education_action_failed` P0	The intended task genuinely fails. Recurring reminder only (permission denied).	`journey`, `failure_reason` = `permission_denied`
`education_action_cancelled` P2	Form-based journeys: the user closed the form. Not counted as a technical failure.	`journey`, `reason` = `form_closed`

4. Activation and setup

Event	When it fires	Parameters
`app_setup_completed` P0	Continue after the selected currency is persisted. Once per install. Setup is not activation.	`setup_path` = `currency_selected`
`home_state_viewed` P0	Home data resolves. Once per foreground visit; resets on background.	`state`, `txn_bucket`
`app_startup_completed` P2	First meaningful screen resolves. Once per launch.	`destination`: `home`, `currency_setup`; `duration_bucket`

5. Scanner and share

Event	When it fires	Parameters
`bill_scan_started` P1	A photo is handed to the scan pipeline.	`source`: `home`, `add_form`, `share_image`
`bill_scan_failed` P1	OCR or extraction throws.	`stage`: `ocr`, `extraction`; `failure_reason`: `invalid_image`, `no_text_found`, `unavailable`, `unexpected`
`bill_scan_preview_shown` P1	Prefilled form is shown from a scan. Once per form.	`result` = `success`, `source`
`bill_scan_abandoned` P1	User closes the scanner without a result, or closes the prefilled form.	`stage`: `capture`, `preview`; `reason`: `closed`, `closed_while_scanning`
`share_text_analyze_start` EXISTING	Shared text analysis begins.	none
`share_text_analyze_result` EXISTING	Shared text analysis ends.	`result`: `empty`, `not_related`, `success`, `failed`

6. Insights, recurring, search, reports

Event	When it fires	Parameters

`insights_viewed` P1	A user-visible Insights state is rendered. Once per distinct state per visit; not while loading.	`state`: `empty`, `ready`, `error`. Error adds `failure_reason` = `persistence`
`insight_action_selected` P1	User acts on a potential-recurring suggestion.	`action_type`: `make_recurring`, `not_recurring`
`recurring_setup_started` P1	Recurring toggled on in the form, or the Insights "Make recurring" sheet opens. Once per form.	`txn_type`, `source`
`recurring_schedule_changed` P1	Recurring edit saved successfully. One event per save.	`change_type`: single field or `multiple`; `changed_fields`: sorted comma-joined `frequency`, `start_date`, `end_date`, `reminder`
`recurring_transaction_paused` P1	Pause swipe action succeeds.	`source` = `recurring_list`
`transaction_search_used` P1	First non-empty debounced query in a search session. Query never sent.	`entry_point` = `home`
`transaction_search_result` P1	Search session ends (search closed); uses last applied query.	`result`: `match`, `no_match`
`report_period_changed` P2	Home month / cycle changes.	`navigation_method`: `arrow`, `swipe`; `direction`: `next`, `previous`

7. Categories, data, settings

Event	When it fires	Parameters
`category_created` P2	Custom category saved successfully.	`category_type`: `expense`, `income`, `savings`
`category_edit` EXISTING	Custom category updated successfully.	none
`export_started` P1	Export begins.	`format` = `json`
`export_completed` P1	Export file produced.	`format` = `json`
`export_failed` P1	Export throws.	`format`, `failure_reason`
`data_import_completed` P1	Backup restored.	`result` = `success`, `record_count_bucket`
`data_import_failed` P1	File picker or restore fails.	`failure_reason`: `permission_denied`, `unsupported_format`, `unexpected`
`clear_data_confirm` EXISTING	User confirms Clear Local Data. Intent, not completion.	none
`local_data_clear_completed` P1	Local deletion succeeds. User properties refresh. Does not reset `first_transaction_saved`.	`source` = `settings`, `result` = `success`
`local_data_clear_failed` P2	Deletion throws.	`failure_reason` = `persistence`
`feature_exposure` EXISTING	The user opens or invokes Insights, Scan, or Export/Import. Not "feature is merely available". Name kept for existing GA4 reports.	`feature_key`: `insights_enabled`, `receipt_scan_enabled`, `export_import_enabled`; `enabled`

8. Notifications and widgets

Event	When it fires	Parameters
`check_in_eligible` P1	Expense check-in eligibility is true. Once per fire date.	`campaign_version`, `trigger_reason` = `inactivity_24h`
`check_in_schedule_result` P1	Local notification scheduling succeeds or throws. Scheduling is not delivery.	`campaign_version`, `result`: `success`, `failed`; failure adds `failure_reason` = `unexpected`
`check_in_cancelled` P1	A pending check-in is removed.	`campaign_version`, `cancellation_reason`: `app_open`, `disabled`, `authorization_denied`, `policy_changed`, `rescheduled`, `inactive`
`check_in_preference_changed` P1	Settings Expense check-in toggle changes.	`enabled`
`check_in_opened` P1	App opened from a notification tap.	`campaign_version` (check-in only), `notification_type`: `expense_check_in`, `recurring_reminder`
`check_in_conversion` P1	A user-created expense is saved within 24 hours of an observed check-in tap, including a later session. Once per marker; a later tap restarts the window. Recurring-generated rows do not convert.	`campaign_version`, `action_type` = `expense_saved`, `window` = `24h`
`widget_destination_opened` P1	App handles a widget deep link. Not an impression.	`widget_type`: `add`, `balance`; `destination`: `home`, `add_transaction`; `widget_family` when known
`widget_transaction_saved` P1	Expense saved in the same session as a Balance-widget check-in tap.	`txn_type`, `source` = `widget_balance`, `widget_family`
`widget_action_failed` P1	Widget add link arrives but the form can't open (not on Home).	`widget_type`, `failure_reason` = `unavailable`

9. App Lock

Event	When it fires	Parameters
`app_lock_enabled` P2	App Lock turned on after authentication.	`source`: `settings`, `deep_link`
`app_lock_disabled` EXISTING	App Lock turned off after authentication.	`source`: `settings`, `deep_link`
`app_lock_auth_result` P2	Any unlock / enable / disable authentication attempt completes.	`result`: `success`, `failed`, `cancelled`; `method`: `face_id`, `touch_id`, `optic_id`, `passcode`; `operation`: `unlock`, `enable`, `disable`
`app_lock_promo_presented` EXISTING	Soft-sell promo shown.	none
`app_lock_promo_enable` EXISTING	Promo primary tapped.	none

app_lock_promo_dismiss EXISTING	Promo dismissed.	none

10. Diagnostics events

Event	When it fires	Parameters
feature_unavailable P2	A feature is hidden by a known constraint. Once per session per feature + reason.	`feature`: `scan`, `insights`, `export_import`; `reason`: `remote_config_off`, `apple_intelligence_unavailable`
slow_operation P2	An operation takes 2 seconds or longer.	`operation`: `scan`, `import`, `export`, `insights_generate`; `duration_bucket`: `2_3s`, `3_10s`, `gt_10s`

11. Review and feedback prompt

Event	When it fires	Parameters
review_eligibility_passed EXISTING	Review prompt becomes eligible. Once per session.	none
review_eligibility_failed EXISTING	Eligibility check fails.	`reason`: `daysSinceInstall`, `launchCount`, `lifetimeTransactions`, `cooldown`, `appStoreOpened`, `criticalErrorThisSession`
review_trigger_txn_milestone EXISTING	Transaction milestone reached.	`milestone`: 20, 50, 100, 250, 500
review_trigger_receipt_milestone EXISTING	Successful scan milestone reached.	`milestone`: 20, 50, 100
review_trigger_insights EXISTING	Insights generated for the first time.	none
review_trigger_seven_day_streak EXISTING	Seven consecutive days of use.	`streak`
review_feedback_flow_presented EXISTING	Review sheet shown.	none
review_positive_selected / **review_negative_selected** EXISTING	"Loving it" or "Have a suggestion" tapped.	none
review_feedback_submitted / **review_feedback_cancelled** EXISTING	Feedback form sent or closed.	none
review_dismissed EXISTING	Review prompt dismissed (starts 30-day cooldown).	none
review_app_store_opened EXISTING	App Store review opened.	none

12. Screens and button clicks

screen_view EXISTING — params: `screen_name`, `screen_class`

`home`, `insights`, `add_transaction`, `edit_transaction`, `manage_categories`, `edit_category`, `settings`, `scan`, `currency_selection`, `currency_list`, `recurring`, `edit_recurring`, `legal_privacy`, `legal_terms`,

`feedback_flow`, `send_feedback`, `app_lock_promo`

`button_click` EXISTING — **params: `button`, `source` (screen)**

`add_transaction`, `open_settings`, `open_insights`, `open_scan`, `scroll_to_top`, `flip_summary_card`, `open_transaction`, `save_transaction`, `delete_transaction`, `scan_bill_from_form`, `take_photo`, `select_library`, `export_backup`, `import_backup`, `manage_categories`, `manage_recurring`, `open_currency`, `open_privacy`, `open_terms`, `send_feedback`, `clear_local_data`, `currency_continue`, `select_currency`, `exclude_recurring_toggle`, `add_category`, `edit_category`, `delete_category`, `pause_recurring`, `skip_recurring`, `edit_recurring`, `delete_recurring`, `save_recurring`, `group_by_date`, `group_by_category`, `open_list_options`, `toggle_group_by_category`, `toggle_show_upcoming_recurring`, `toggle_filter_expense`, `toggle_filter_income`, `toggle_filter_savings`, `set_sort_ascending`, `set_sort_descending`, `open_upcoming_recurring`, `search_transactions`, `review_loving_it`, `review_have_suggestion`, `review_leave_review`, `review_send_feedback`, `review_dismiss`, `app_lock_promo_enable`, `app_lock_promo_not_now`, `app_lock_toggle`

Firebase automatic events are also available: `first_open`, `session_start`, `user_engagement`, `app_remove` (unreliable on iOS).

13. User properties

Refreshed when the app becomes active and after transaction, recurring, import, App Lock, or check-in changes (debounced 300 ms). User ID is the anonymous identity, set via `setUserID`; no unique ID is stored as a user property.

Property	Values	Derived from
`has_expense`	`true`, `false`	Any expense transaction exists
`has_income`	`true`, `false`	Any income transaction exists
`has_recurring`	`true`, `false`	Any active recurring rule
`txn_bucket`	`0`, `1`, `2_5`, `6_20`, `21_plus`	Lifetime transaction count
`notifications_authorized`	`yes`, `no`, `provisional`, `unknown`	System notification authorization status
`app_lock_enabled`	`true`, `false`	App Lock setting
`expense_check_in`	`enabled`, `disabled`	Expense check-in setting

14. Crashlytics non-fatals

Recorded through `AppDiagnostics.recordNonFatal` for unexpected, actionable failures. Each record is a sanitized `NSError` with domain `com.snapledger.diagnostics.<operation>` and a code derived from the failure reason. `userInfo` holds only `operation`, `feature`, `reason`, `underlying_domain`, and `underlying_code`, never the error description or any payload.

Operation	Feature	Recorded when
`transaction_add`	`transactions`	SwiftData insert fails
`transaction_update`	`transactions`	SwiftData update fails
`transaction_delete`	`transactions`	SwiftData delete fails
`recurring_create`	`recurring`	Recurring rule create fails
`recurring_update`	`recurring`	Recurring rule update fails
`recurring_pause`	`recurring`	Recurring pause / cancel fails
`category_save`	`categories`	Category save fails for a non-validation reason

export	import_export	Export throws
import	import_export	Import fails for an unexpected reason (not bad format or permission)
scan	scanner	Scan pipeline throws a non-scanner error
insights_load	insights	Insights data fetch fails
home_load	home	Home data or lifetime-count fetch fails
local_data_clear	settings	Clear Local Data throws

Debug builds also print `[diagnostics] non-fatal <operation> feature=… reason=…` and `[analytics] <event> keys=[…]` to the console.

15. Renamed or removed in this rollout

Hard rename: the old names are no longer sent. Historical data under the old names stays in Firebase.

Old event	Replaced by
txn_delete	transaction_deleted
receipt_scan_start	bill_scan_started
receipt_scan_result	bill_scan_failed ; success is bill_scan_preview_shown
category_add	category_created
export_backup (event)	export_started / export_completed / export_failed
import_backup (event)	data_import_completed / data_import_failed
expense_reminder_eligible	check_in_eligible
expense_reminder_scheduled	check_in_schedule_result
expense_reminder_cancelled	check_in_cancelled
expense_reminder_opened	check_in_opened
expense_reminder_expense_added	check_in_conversion
widget_add_expense_opened	widget_destination_opened
widget_expense_saved	widget_transaction_saved
button_click **month_previous** / **month_next**	report_period_changed

share_text_analyze_result now sends failed instead of fail .

16. Inactivity and reactivation (report definitions, no extra events)

Do not add a custom foreground event. Use Firebase session_start plus successful meaningful actions. Retention is by analytics identity / this install, not a verified person.

Milestone	Definition
Activated	At least one first_transaction_saved
Repeat user	A later session includes txn_save , recurring create, export_completed , or insights_viewed state=ready
At risk	No meaningful action for 7 days (GA4 cohort)
Inactive	No meaningful action for 14 or 30 days (inactivity proxy, not person-level churn)

Reactivated	After 14+ days without a meaningful action, returns and completes one (cohort/query; no client event)

17. Not emitted

- **Widget refresh result:** WidgetKit does not report reload success, so there is nothing to observe.
- **Notification display and widget impressions:** iOS does not expose these reliably, so they are not claimed.
- **Validation for empty amount or description:** Save stays disabled, so no failed attempt occurs.
- **feature_used:** not added. `feature_exposure` already means the feature was opened/invoked; renaming would break existing GA4 reports.
- **Client churn / reactivated events:** defined in GA4, not emitted by the app.